

Predicting Public Opinion with MRP

In this project I use an MRP to model attitudes towards economic growth and environmental protection. The dependent variable is an ordinal survey question from the British Election Study 2024 that asks respondents to indicate which of the two objectives should take priority. I use Census 2021 data to infer the mean response by region.

About the data

I use two datasets in this project: Census 2021 microdata (teaching sample), published by the Office for National Statistics (ONS) and the 2024 British Election Study internet panel data, published by the British Election Study (BES).

Census microdata tibble:

```
# A tibble: 604,351 × 19
  resident_id_m approx_social_grade country_of_birth_3a
economic_activity_status_10m
  <chr>                <int>                <int>
<int>
1 PTS000000588097      4                  1
1
2 PTS00000000320      -8                  1
5
3 PTS000000397448      -8                  2
5
4 PTS000000082442      -8                  1
5
5 PTS000000016066      4                  1
8
6 PTS000000294257      -8                  2
5
7 PTS000000356470      3                  1
2
8 PTS000000259832      4                  1
-8
9 PTS000000007691      3                  1
```

```

1
10 PTS000000348026          4          2
1
# i 604,341 more rows
# i 15 more variables: ethnic_group_tb_6a <int>, health_in_general <int>,
#   hh_families_type_6a <int>, hours_per_week_worked <int>,
in_full_time_education <int>,
#   industry_10a <int>, iol22cd <chr>, legal_partnership_status_6a <int>,
occupation_10a <int>,
#   region <chr>, religion_tb <int>, residence_type <int>, resident_age_7d
<int>, sex <int>,
#   usual_short_student <int>

```

The Census microdata teaching sample is a publicly available dataset that consists of a random sample of 1% of the data in the full Census.

While this dataset contains valuable and informative demographic variables, such as ethnicity and sex, it also lacks data on other important variables such as education and own/rent status. These omitted variables are the kind that would likely correlate quite strongly with the dependent variable, meaning that it might be difficult to build a model with lots of explanatory power. Fortunately, a model like this is unlikely to be essential for a good result with this project.

British Election Study tibble:

```

# A tibble: 3,089 × 373
  finalserialno Y10A Y10B1 Y10B2 Y10B3 Y10B4 Y10B5 a01 a01_code
a02 a03
  <dbl> <dbl> <dbl+lbl> <dbl+1> <dbl+1> <dbl+1> <dbl+1> <chr> <dbl+lb>
<dbl+lb> <dbl+1>
1 101103 3 1 [Yes] 0 [No] 0 [No] 0 [No] 0 [No] Cost... 12
[Liv... 1 [Lab... 2 [Fai...
2 101112 NA NA NA NA NA NA lack... 20
[Poo... 0 [Non... 4 [Not...
3 101201 NA NA NA NA NA NA ineq... 11
[Ine... 1 [Lab... 2 [Fai...
4 101202 NA NA NA NA NA NA safe... 5
[Def... 1 [Lab... 2 [Fai...
5 101203 2 1 [Yes] 0 [No] 0 [No] 0 [No] 0 [No] Immi... 10
[Imm... -1 [Don... 3 [Not...
6 101208 NA NA NA NA NA NA cost... 12
[Liv... 0 [Non... 3 [Not...
7 101212 NA NA NA NA NA NA immi... 10
[Imm... 0 [Non... 2 [Fai...
8 101213 NA NA NA NA NA NA nast... 4
[Cri... -1 [Don... 4 [Not...

```

```

  9      102104    NA NA      NA      NA      NA      NA      nati... 17
[Nat... 1 [Lab... 3 [Not...
10      102105    NA NA      NA      NA      NA      NA      immi... 10
[Imm... 8 [Ref... 2 [Fai...
# i 3,079 more rows
# i 362 more variables: m02_1 <dbl+lbl>, m02_2 <dbl+lbl>, m02_3 <dbl+lbl>, m02_4
<dbl+lbl>,
#   m02_5 <dbl+lbl>, m02_6 <dbl+lbl>, b01 <dbl+lbl>, b02 <dbl+lbl>, b04
<dbl+lbl>,
#   b05 <dbl+lbl>, b0601 <dbl+lbl>, b0602 <dbl+lbl>, b0603 <dbl+lbl>, b0604
<dbl+lbl>,
#   b0605 <dbl+lbl>, b0606 <dbl+lbl>, b06a <dbl+lbl>, b07 <dbl+lbl>, b09
<dbl+lbl>,
#   b09b <dbl+lbl>, b12_1 <dbl+lbl>, b12_2 <dbl+lbl>, b12_3 <dbl+lbl>, b12_4
<dbl+lbl>,
#   b12_5 <dbl+lbl>, b12_6 <dbl+lbl>, b13_1 <dbl+lbl>, b13_2 <dbl+lbl>, b13_3
<dbl+lbl>, ...

```

Although the raw variable names are cryptic, it's immediately apparent that the BES internet panel data is on the other end of the scale to the Census micro data. Unlike the Census microdata, the full BES internet panel data is publicly available, offering data on public opinion across a huge array of topics (+ demographics) measured over 372 variables.

Data Cleaning and Preparation

```

#Load libraries
library(tidyverse)
library(haven)
library(naniar)
library(brms)
library(sf)

#For reproducibility
set.seed(1234)

#Load in data
Census <- read.csv("D:\\Datasets\\Census_Microdata.csv")
head(Census)
tibble(Census)

#Check for missing values
sum(is.na(Census))

```

Out:


```

cleanBES$Ethnicity == 5 ~ 3,
cleanBES$Ethnicity == 6 ~ 3,
cleanBES$Ethnicity == 7 ~ 3,
cleanBES$Ethnicity == 8 ~ 3,
cleanBES$Ethnicity == 9 ~ 1,
cleanBES$Ethnicity == 10 ~ 1,
cleanBES$Ethnicity == 11 ~ 1,
cleanBES$Ethnicity == 12 ~ 1,
cleanBES$Ethnicity == 13 ~ 1,
cleanBES$Ethnicity == 14 ~ 2,
cleanBES$Ethnicity == 15 ~ 2,
cleanBES$Ethnicity == 16 ~ 2,
cleanBES$Ethnicity == 17 ~ 5,
cleanBES$Ethnicity == 18 ~ 5,
cleanBES$Ethnicity == 19 ~ 5,
cleanBES$Ethnicity == -999 ~ -8,
cleanBES$Ethnicity == -2 ~ -8)

cleanBES$Age <- as.numeric(cleanBES$Age)

cleanBES$Age <- case_when(
  between(cleanBES$Age, 1, 15) ~ 1,
  between(cleanBES$Age, 16, 24) ~ 2,
  between(cleanBES$Age, 25, 34) ~ 3,
  between(cleanBES$Age, 35, 44) ~ 4,
  between(cleanBES$Age, 45, 54) ~ 5,
  between(cleanBES$Age, 55, 64) ~ 6,
  cleanBES$Age >= 65 ~ 7,
  cleanBES$Age == -2 ~ NA
)

cleanBES <- na.omit(cleanBES)

unique(cleanBES$Age)

unique(cleanBES$Sex)
unique(Census$sex)
sum(cleanBES$Sex == -999)

cleanBES$Sex <- case_when(
  cleanBES$Sex == -999 ~ -8,
  cleanBES$Sex == 1 ~ 1,
  cleanBES$Sex == 2 ~ 2,
  cleanBES$Sex == 3 ~ 1,
  cleanBES$Sex == 4 ~ 2
)

unique(cleanBES$Sex)

unique(cleanBES$Region)
unique(Census$region)

```

```

cleanBES$Region <- case_when(
  cleanBES$Region == 1 ~ 'E12000004',
  cleanBES$Region == 2 ~ 'E12000006',
  cleanBES$Region == 3 ~ 'E12000007',
  cleanBES$Region == 4 ~ 'E12000001',
  cleanBES$Region == 5 ~ 'E12000002',
  cleanBES$Region == 6 ~ 'S99999999',
  cleanBES$Region == 7 ~ 'E12000008',
  cleanBES$Region == 8 ~ 'E12000009',
  cleanBES$Region == 9 ~ 'W92000004',
  cleanBES$Region == 10 ~ 'E12000005',
  cleanBES$Region == 11 ~ 'E12000003',
)

```

```

unique(Census$religion_tb)
unique(cleanBES$Religion)

```

```

cleanBES$Religion <- case_when(
  cleanBES$Religion == -999 ~ -8,
  cleanBES$Religion == -2 ~ 9,
  cleanBES$Religion == 0 ~ -8,
  cleanBES$Religion == 1 ~ 2,
  cleanBES$Religion == 2 ~ 2,
  cleanBES$Religion == 3 ~ 2,
  cleanBES$Religion == 4 ~ 2,
  cleanBES$Religion == 5 ~ 2,
  cleanBES$Religion == 6 ~ 2,
  cleanBES$Religion == 7 ~ 2,
  cleanBES$Religion == 8 ~ 2,
  cleanBES$Religion == 9 ~ 2,
  cleanBES$Religion == 10 ~ 2,
  cleanBES$Religion == 11 ~ 2,
  cleanBES$Religion == 12 ~ 5,
  cleanBES$Religion == 13 ~ 4,
  cleanBES$Religion == 14 ~ 6,
  cleanBES$Religion == 15 ~ 7,
  cleanBES$Religion == 16 ~ 3,
  cleanBES$Religion == 17 ~ 8
)

```

```

cleanBES$Employment <- case_when(
  cleanBES$Employment == 1 ~ 1,
  cleanBES$Employment == 2 ~ 2,
  cleanBES$Employment == 3 ~ 1,
  cleanBES$Employment == 4 ~ 2,
  cleanBES$Employment == 5 ~ 3,
  cleanBES$Employment == 6 ~ 9,
  cleanBES$Employment == 7 ~ 6,
  cleanBES$Employment == 8 ~ 7,
  cleanBES$Employment == 9 ~ 9,
)

```

```

cleanBES$Employment == 10 ~ 8,
cleanBES$Employment == 11 ~ 5,
cleanBES$Employment == 12 ~ 9,
cleanBES$Employment == -999 ~ -8
)

unique(cleanBES$MaritalStatus)
unique(Census$y26)

cleanBES$MaritalStatus <- case_when(
  cleanBES$MaritalStatus == 1 ~ 2,
  cleanBES$MaritalStatus == 2 ~ 1,
  cleanBES$MaritalStatus == 3 ~ 1,
  cleanBES$MaritalStatus == 4 ~ 5,
  cleanBES$MaritalStatus == 5 ~ 3,
  cleanBES$MaritalStatus == 6 ~ 4,
  cleanBES$MaritalStatus == -999 ~ -8
)

#Take a subset of the full census data so that only relevant variables are kept
cleanCensus <- subset(Census, select = c(resident_id_m, resident_age_7d, sex,
ethnic_group_tb_6a, region, religion_tb, economic_activity_status_10m,
legal_partnership_status_6a))

cleanCensus <- rename(cleanCensus, "ID" = resident_id_m, "Age" =
resident_age_7d, "Sex" = sex, "Ethnicity" = ethnic_group_tb_6a, "Region" =
region, "Religion" = religion_tb, "Employment" = economic_activity_status_10m,
"MaritalStatus" = legal_partnership_status_6a)

#Ensure that age is numeric so it can be parsed correctly by the model
cleanCensus$Age <- as.numeric(cleanCensus$Age)

```

While many of the response categories matched without any need for substantive modifications, some were not perfectly aligned. For instance, the BES, which is far more granular than the Census, allowed respondents to indicate whether they work full time or part time, whereas the Census did not split response categories by hours at all. This meant that multiple categories from the BES had to be combined into one category, such that it would be compatible with the Census. This likely won't influence the effectiveness of the model too much, but it's one of the trade-offs that MRP workflows often involve.

Fitting the Model

Now that the data has been prepared, we can use it to fit a model. I'm using the 'brms' package to fit a Bayesian ordinal regression model, which is standard practice for MRPs that involve ordinal data. Unlike frequentist models, Bayesian models output a distribution of possible coefficients rather than a point estimate, making them particularly informative given the sample size we are working with.

```
#Ensure dependent variable is parsed as ordinal
cleanBES$EnvironmentEconomy <- ordered(as.factor(cleanBES$EnvironmentEconomy))

#Fit model to cleanBES
Model <- brm(formula = EnvironmentEconomy ~ Sex + Ethnicity + Religion +
  Employment + MaritalStatus + (1 | Region:Age) + (1 | Region),
  data = cleanBES,
  family = cumulative("logit"), #For ordinal data
  chains = 4,
  cores = 4, #Split workload across multiple cores for speed
  iter = 2000,
  backend = 'cmdstanr'
)
```

```
summary(Model)
```

Out:

```
Family: cumulative
Links: mu = logit
Formula: EnvironmentEconomy ~ Sex + Ethnicity + Religion + Employment +
MaritalStatus + (1 | Region:Age) + (1 | Region)
Data: cleanBES (Number of observations: 2904)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 4000

Multilevel Hyperparameters:
~Region (Number of levels: 11)
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.24    0.09    0.11    0.45 1.00    1466    1401

~Region:Age (Number of levels: 66)
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.15    0.06    0.03    0.27 1.01     935    1547

Regression Coefficients:
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept[1]   -2.90    0.21   -3.30   -2.48 1.00    4762    2824
Intercept[2]   -2.29    0.20   -2.67   -1.89 1.00    4990    2949
```

Intercept[3]	-2.08	0.20	-2.45	-1.68	1.00	5067	3092
Intercept[4]	-1.62	0.19	-1.99	-1.23	1.00	5015	3297
Intercept[5]	-1.17	0.19	-1.54	-0.78	1.00	5112	3269
Intercept[6]	-0.79	0.19	-1.17	-0.41	1.00	5174	3037
Intercept[7]	0.34	0.19	-0.03	0.72	1.00	5297	3243
Intercept[8]	0.77	0.19	0.40	1.15	1.00	5327	3127
Intercept[9]	1.35	0.19	0.99	1.74	1.00	5268	3349
Intercept[10]	2.19	0.20	1.80	2.58	1.00	5146	2977
Intercept[11]	2.66	0.20	2.27	3.05	1.00	5156	3213
Sex	0.22	0.07	0.09	0.35	1.00	6208	3023
Ethnicity	0.03	0.04	-0.04	0.10	1.00	7071	3190
Religion	-0.02	0.01	-0.03	-0.01	1.00	5990	3497
Employment	-0.00	0.01	-0.03	0.02	1.00	5261	2887
MaritalStatus	-0.04	0.03	-0.09	0.02	1.00	5849	3100

Further Distributional Parameters:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
disc	1.00	0.00	1.00	1.00	NA	NA	NA

Draws were sampled using `sample(hmc)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat = 1`).

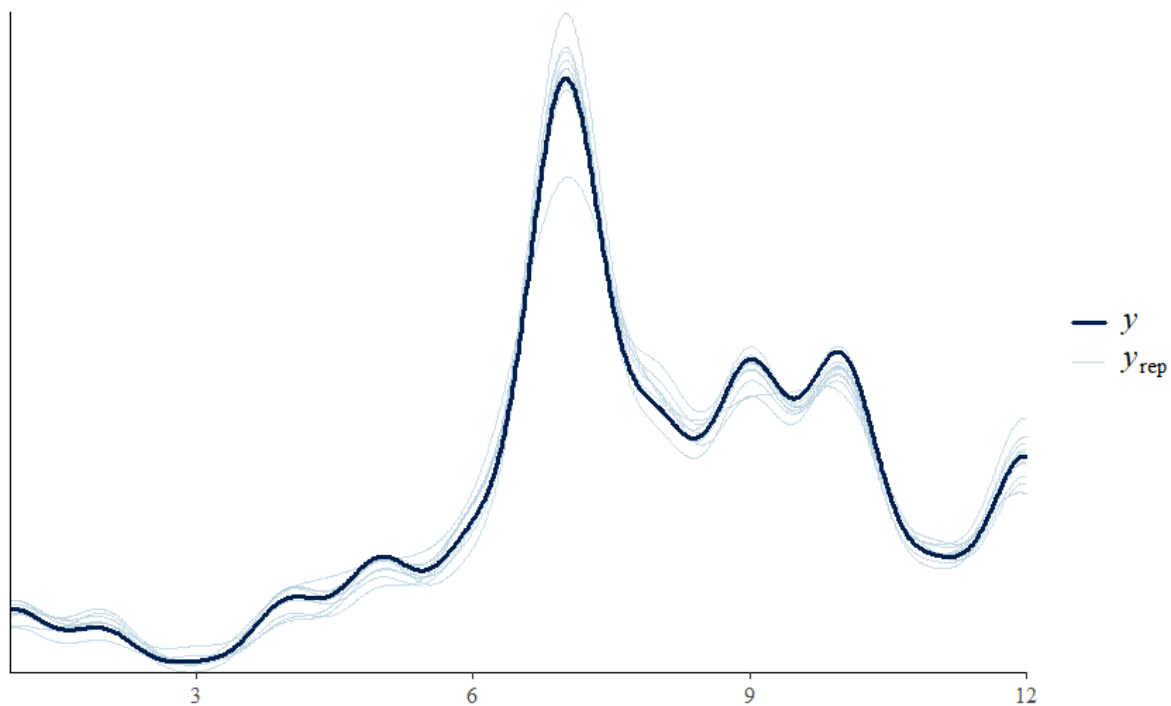
The regression table for the first model shows that only sex and, to a lesser extent, religion might credibly correlate with the dependent variable. Again, other variables such as education are likely to have been helpful here for more explanatory power.

`Rhat` is 1.00 for each of the predictors, indicating that the chains converged well.

We can run a graphical posterior predictive check to evaluate how well the model fits the data. To do this, the check uses the model to simulate new data and compares it with the real data.

```
pp_check(Model)
```

Out:



Here, y represents the actual data, y_{rep} represents data that the model simulated. The plot indicates that the model seems to fit the data quite well, with the troughs and peaks mostly lining up. This suggests that the model is likely to perform predictive tasks well.

Post-stratification

Now that we've fit the model, we can use it to predict the dependent variable from the Census microdata.

To do this, we need a post-stratification frame. This is a dataset that contains every unique configuration of variables within the full Census microdataset without duplicates. This means that the model runs on one of each configuration of variables in the dataset, avoiding having to perform the same calculations over and over again. This makes the modelling process far less compute-expensive and quicker.

```
#Create post-stratification frame
```

```

Frame <- cleanCensus %>%
  count(Sex, Ethnicity, Religion, Employment, MaritalStatus, Region, Age, name =
    "n") %>%
  as.data.frame()

nrow(cleanCensus)
nrow(Frame)

#Predict on the post-stratification frame
Poststrat <- posterior_epred(
  Model,
  newdata = Frame,
  ndraws = 1000,
  allow_new_levels = TRUE
)

#Use these predictions to apply a weighted mean across the
nCategories <- dim(Poststrat)[3] #Find how many categories there are
categoryVals <- as.numeric(dimnames(Poststrat)[[3]]) #Find the numeric label of
the category (likert scale score)

predScore <- apply(Poststrat, c(1, 2), function(p) sum(p * categoryVals))
#Multiply each category's probability by numeric value and sum them
#predScore contains a single score rather than a probability vector

#Convert to dpylr compatible object
predLong <- as.data.frame.table(predScore, responseName = "score")
names(predLong)[1:2] <- c("draw", "cell")
predLong <- predLong %>% mutate(cell = as.integer(cell))

#Find which region each cell corresponds to
cellSearch <- Frame %>% mutate(cell = row_number()) %>% select(cell, Region, n)

#Add the result to predLong
predLong <- predLong %>% left_join(cellSearch, by = "cell")

#Calculate a population-weighted average per region, per draw
regionDraws <- predLong %>% group_by(Region, draw) %>% summarise(WeightedScore =
sum(score * n) / sum(n), .groups = "drop")

#Take a single number from the weighted average probability distribution, and
construct 95% credible intervals
regionSummary <- regionDraws %>% group_by(Region) %>% summarise(
  mean = mean(WeightedScore),
  lower = quantile(WeightedScore, 0.025),
  upper = quantile(WeightedScore, 0.975),
  .groups = "drop"
)

```

Now that we've finished the MRP, let's visualise it as a choropleth!

```
#Join the geojson data to the post-stratification data
mapData <- SF %>% left_join(regionSummary, by = c("eer16cd" = "Region"))

sum(is.na(mapData$mean))
#1 NA - Scotland

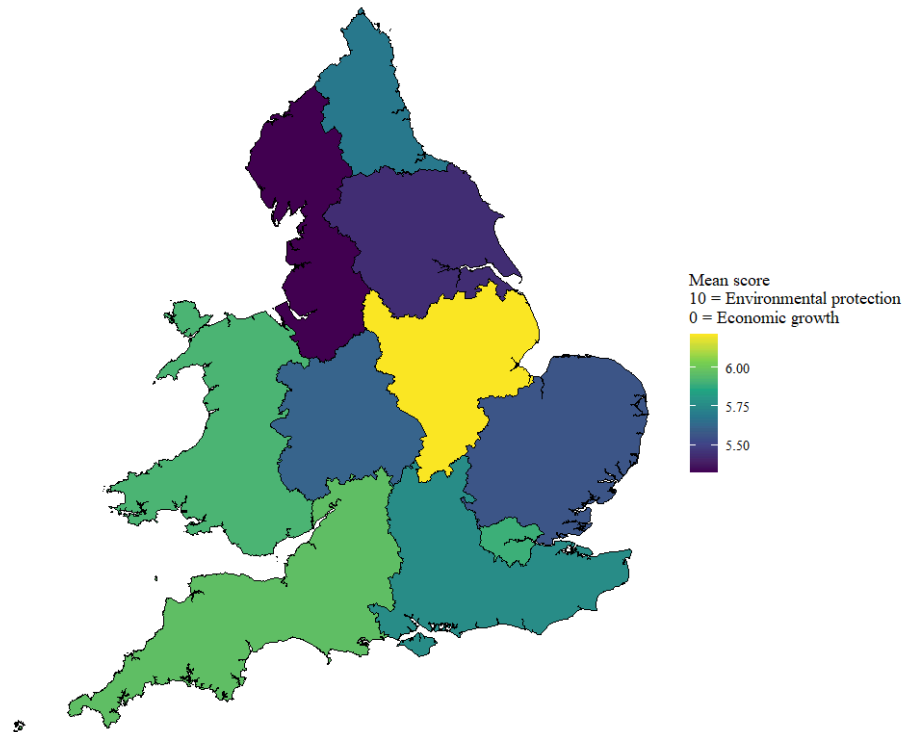
#Remove NA (Scotland isn't included because it's Census was conducted
separately)
mapData <- na.omit(mapData)

#Create the plot
choropleth <- ggplot(data = mapData) +
  geom_sf(aes(fill = mean), color = "black", linewidth = 0.3) +
  scale_fill_viridis_c(name = "Mean score\n10 = Environmental protection\n0 =
Economic growth") +
  labs(
    title = "Should economic growth or environmental protection take priority?",
    caption = "Regional MRP modelled on Census 2021 data and British Election
Study 2024 data."
  ) +
  theme_void() +
  theme(text = element_text(family = "serif")) +
  theme(plot.title = element_text(hjust = 0.5, size = 16))

#Visualise the plot
choropleth
```

Out:

Should economic growth or environmental protection take priority?



Regional MRP modelled on Census 2021 data and British Election Study 2024 data.

The choropleth shows that the by-region variation in opinion on this subject is fairly narrow, with most regions falling somewhere in the middle, between 5 and 6.5. These findings suggest that the British public are generally interested in striking a balance between economic growth and environmental protection, rather than simply prioritising one over the other. This analysis has shown that other demographic variables, such as sex, may explain variation in opinion on this issue at least as much as region.

As a speculative aside, it seems that regions in the north east and west appear to be slightly more in favour of economic growth than others, perhaps reflecting a greater prevalence of materialist concerns than the wealthier southern regions, where environmental protection is slightly preferred. While our analysis cannot be used to support this hypothesis, it does indicate that research into the nature of this phenomenon might yield interesting results.

